Q&A

THINGS YOU WANTED TO KNOW!

Ques. WHAT ARE THE COMPONENTS USED IN DRONE DESIGN? PLEASE SUGGEST SOME DRONE PROJECTS USING RASPBERRY PI OR ARDUINO, AND THEIR AVAILABILITY.

Pamarthi Kanakaraja

Ans. An unmanned aerial vehicle (UAV) or drone is an aircraft without a human pilot aboard. Components required for its design depend on type. There are different types of drones depending on technology and application. Applications include civil, commercial and military. While Chinese DJI is a leading manufacturer of commercial and civilian drones, many other people/companies around the world are also developing drones. Hence, there is no comprehensive list of drone systems.

Most common drones have two units: transmitter and receiver. The receiver unit consists of the drone, and the transmitter unit is normally a ground-based controller system using which the user can send commands (operations to be performed) to the drone.

The transmitter is first calibrated and tested using commands, which the drone has to perform. There may be drones that do not require the traditional remote transmitter unit. Flight of the drone may operate with various degrees of autonomy, either under remote control of a human operator or autonomously by onboard computer.

Given below is a list of common components used in drone design:

Body and chassis. There are mainly three types of drones: tail-less multi-copter, tailed mono- and bi-copters, and fixed wings. Quadcopters are the most popular multi-copter drones. Drone body shells for these are available in various online stores.

Power supplies. There are different power sources for a drone, including battery, solar, hydro fuel cell, combustion engine, cable-tethered power supply and laser beam power. The most common source of power is a lithium-polymer battery.

Actuators and motors. These include digital electronic speed controllers linked to engines, propellers, servomotors and others.

Sensors. Various sensors are used in a drone as per requirement and application. Position and movement sensors give information about the state of the aircraft. Common sensors are laser, radar, camera, gyroscope, accelerometer, compass, barometer and GPS receiver.

Computing. Drones use advanced computing technology, from analogue controls to microcontrollers, systems-on-chip and single-board computers.

Software. Flight stack or autopilot software are used in drone applications. Open source software like ArduPilot, OpenPilot, Paparazz and others are easily available.

Also, there are many smartphone apps available for drones.

Remote control. Most drones use radio frequency to communicate between the remote and the drone, which may be capable of autonomous or semi-autonomous operations. Remote control signals from the operator side can be issued from one of the following:

- Ground control, which is a human operating a radio transmitter, smartphone, computer or other similar control system.
- Remote network system, like satellite duplex data links, cellular mesh and LTE network.
- Another aircraft serving as a relay or mobile control station.

There are many websites for hobbyists to start with drone design using Raspberry Pi and Arduino. For Raspberry Pi, you may refer to *www.instructables.com/id/The-Drone-Pi/*

For Arduino, you may check out *www.instructables.com/id/How-to-Make-a-Drone-Using-Arduino-Make-a-Quadcopte/*

You can buy components for drones from online stores such as *www.amazon.in*, *www.banggood.com* and *www.robokits.co.in*

Q2. WHAT IS THE DIFFERENCE BETWEEN ADSL AND VDSL?

A. Samiuddhin

A2. Digital subscriber line (DSL) technology offers high-speed Internet service for homes and offices over telephone lines. It is unique because you can use the Internet and make telephone calls at the same time. A DSL system separates telephone signals into three frequencies bands: lowest band for telephone calls, and other two for uploading and downloading online activities.

There are two types of DSL: asymmetric digital subscriber line (ADSL) and symmetric digital subscriber line (SDSL). If data throughput in upstream direction (towards service provider) is lower, it is ADSL. In SDSL service, both downstream and upstream data rates are equal.

Bit rate of DSL services (downstream) typically ranges from 256kbps to over 100Mbps, depending on DSL technology, line conditions and service-level implementation.

Very-high-bit-rate digital subscriber line (VDSL) and very-high-bit-rate digital subscriber line 2 (VDSL2) technologies provide data transmission faster than ADSL. These can also be deployed over existing wires used for analogue telephone service.

VDSL offers speeds of up to 52Mbps downstream and 16Mbps upstream, and is capable of supporting applications such as high-definition television (HDTV), telephone services (voice over IP) and general Internet access, over a single connection.

VDSL2 (second-generation system) uses frequencies up to 30MHz to provide data rates exceeding 100Mbps, simultaneously in both upstream and downstream directions.

ADSL technology covers a larger distance than VDSL. It supports only asymmetric data and plain old telephone (POT) service, while VDSL supports asymmetric, symmetric data and POTs.

Answers compiled by Sani Theo, technical editor. Letters and questions for publication may be addressed to Editor, Electronics For You, D-87/1, Okhla Industrial Area, Phase 1, New Delhi 110020 (e-mail: **editsec@efy.in**) and should include name and address of the sender.